

**MACHINE LEARNING-BASED STUDENT PERFORMANCE AND
DROPOUT RISK PREDICTION FOR EARLY IDENTIFICATION OF AT-
RISK STUDENTS**

PHASE I REPORT

Submitted by

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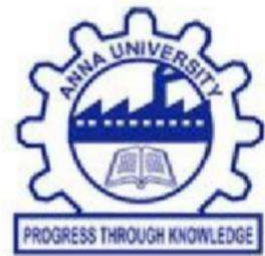
in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING



**RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI
ANNA UNIVERSITY: CHENNAI 600 025**

September 2026

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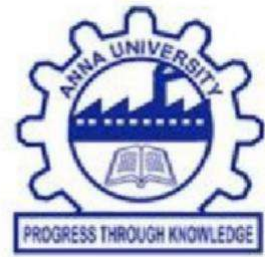
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BONAFIDE CERTIFICATE

Certified that this Report titled “**MACHINE LEARNING-BASED STUDENT PERFORMANCE AND DROPOUT RISK PREDICTION FOR EARLY IDENTIFICATION OF AT-RISK STUDENTS**” is the Bonafide work of **Musa Haji (230701391), Prithivi Raj (230701508) and Sachin Chaudhary (230701402)** who

carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

This project addresses the following Sustainable Development Goal(s):

Goal No: 4

Description: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

Goal No: 10

Description: Reduce inequality within and among countries by identifying students who are at academic risk and supporting timely interventions to improve educational outcomes.

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ABSTRACT

The increasing availability of student academic and behavioral data provides an opportunity to identify students who may be at academic risk at an early stage. Traditional methods of evaluating student performance often rely on manual assessment of grades and attendance, which may not effectively capture complex patterns across multiple academic, demographic, and behavioral factors. To address this challenge, this project proposes a Machine Learning-Based Student Performance and Dropout Risk Prediction System for the early identification of at-risk students. The proposed system uses student academic, demographic, social, and behavioral attributes to analyze performance patterns and classify students into different performance categories. The system begins with data preprocessing, including handling missing values, removing inconsistent records, encoding categorical attributes, transforming numerical features, and preparing the dataset for machine learning. Multiple classification algorithms, including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, Support Vector Machine, and Gradient Boosting, are trained and evaluated for student performance prediction. Stratified cross-validation and hyperparameter tuning are employed to improve model reliability and identify the most suitable predictive model. Based on the experimental evaluation, the tuned Gradient Boosting model achieved an accuracy of 87.34%, with a precision of 87.95%, recall of 87.34%, and F1-score of 87.42% on the test dataset. Feature-importance analysis further indicates that previous academic performance, particularly the G2 grade, has a major influence on the prediction outcome, while other factors such as G1, age, attendance-related behavior, alcohol consumption, parental occupation, and past failures also contribute to the classification. The system can classify students into Low, Average, and Good performance categories and provide prediction probabilities for individual students, supporting early identification and targeted academic intervention. Although the current dataset does not contain a dedicated dropout-labelled target, the developed performance prediction framework establishes the foundation for an extended dropout-risk prediction module. Future development will incorporate appropriately labelled dropout data to enable direct dropout-risk classification and more comprehensive early-warning support for educational institutions.

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LIST OF ABBREVIATIONS

ML	Machine Learning
AI	Artificial Intelligence
XAI	Explainable Artificial Intelligence
SHAP	SHapley Additive exPlanations
SVM	Support Vector Machine
KNN	K-Nearest Neighbors
CGPA	Cumulative Grade Point Average
UCI	University of California, Irvine
DFD	Data Flow Diagram
CSV	Comma-Separated Values
CPU	Central Processing Unit
GPU	Graphics Processing Unit
RAM	Random Access Memory
SSD	Solid State Drive
GB	Gigabyte
ROC-AUC	Receiver Operating Characteristic – Area Under Curve
API	Application Programming Interface

CHAPTER 1

INTRODUCTION

1.1 General

The increasing availability of educational data has created new opportunities to understand and improve student academic performance. Educational institutions collect information related to students' academic records, attendance, demographics, family background, social activities, and other behavioral factors. Analysing these factors manually can be time-consuming and may not always reveal the patterns that indicate whether a student is likely to perform well or require additional academic support.

Student performance can be influenced by multiple factors, and identifying students who are struggling at an early stage is important for providing timely guidance and intervention. Traditional evaluation methods generally focus on examination marks and periodic assessments, which may not be sufficient to identify underlying patterns in student performance. Machine learning techniques can help analyse historical student data and discover relationships between different attributes and academic outcomes.

The proposed Machine Learning-Based Student Performance and Dropout Risk Prediction System uses machine learning algorithms to analyse student-related data and predict academic performance. The system preprocesses the collected dataset by handling missing or inconsistent data, encoding categorical attributes, transforming numerical features, and preparing the data for model training. Different machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, Support Vector Machine, and Gradient Boosting are evaluated to determine their effectiveness in student performance classification. The system classifies students into Low, Average, and Good performance categories and provides prediction probabilities for individual students. Feature-importance analysis is also used to understand which factors have a greater influence on the prediction, making the system more interpretable and useful for academic decision-making. The developed framework provides a foundation for the early identification of students who may require additional academic attention. Furthermore, the system can be extended with a suitable dropout-labelled dataset to directly predict dropout risk and support educational institutions in developing timely intervention strategies.

1.2 Objective

The main aim of the proposed system is to develop a machine learning-based student performance and dropout risk prediction system that can analyse student-related academic, demographic, social, and behavioural factors to identify students who may require early academic support.

The specific objectives are:

1. To collect and analyse student academic, demographic, social, and behavioural data.
2. To preprocess the student dataset by handling missing values, inconsistent records, and duplicate data.
3. To encode categorical attributes and transform numerical features into a suitable format for machine learning.
4. To divide the prepared dataset into training and testing sets for model development and evaluation.
5. To identify important factors that influence student academic performance.
6. To train multiple machine learning classification algorithms for predicting student performance.
7. To evaluate models using Accuracy, Precision, Recall, and F1-score.
8. To use stratified cross-validation to assess the reliability and consistency of the trained models.
9. To perform hyperparameter tuning to improve the performance of suitable machine learning models.
10. To classify students into Low, Average, and Good academic performance categories.
11. To generate prediction probabilities for individual students to provide a better understanding of the model's prediction.
12. To apply feature-importance analysis for understanding the factors contributing to student performance predictions.

13. To provide an early identification mechanism for students who may require additional academic guidance and intervention.
14. To establish a foundation for extending the system with a dedicated dropout-risk prediction module using an appropriately labelled dropout dataset.

Implementation Objective

For the present project implementation, the scope is restricted to:

- Student Data Collection and Preprocessing
- Categorical Encoding and Numerical Feature Transformation
- Student Performance Classification
- Machine Learning Model Training and Comparison
- Cross-Validation and Hyperparameter Tuning
- Performance Evaluation using Accuracy, Precision, Recall, and F1-score
- Feature-Importance Analysis
- Individual Student Performance Prediction
- Prediction Probability Generation
- Early Identification of Students Requiring Academic Support

1.3 Existing System

Multiple approaches and machine learning models have been developed for analysing and predicting student academic performance. Traditional educational systems generally depend on examination marks, attendance records, assignments and periodic assessments to evaluate students. Statistical techniques and conventional classification algorithms are also used to analyse student-related data and identify factors that influence academic outcomes. Datasets such as the Student Performance dataset provide academic, demographic, social and behavioural attributes that can be used for predicting student performance.

Apart from traditional statistical approaches, various machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, Support Vector

Machine and Gradient Boosting have been applied to student performance prediction. These models can identify patterns within historical student records and classify students according to their expected academic performance. However, the effectiveness of these approaches depends heavily on the quality of the dataset, appropriate preprocessing, feature selection and the distribution of the target classes. In cases where some performance categories contain very few students, class imbalance can negatively affect the prediction results.

The major limitation of existing student performance assessment systems is that they primarily focus on analysing academic results after the completion of assessments rather than providing an early indication of students who may require support. Manual analysis of multiple academic, demographic and behavioural factors can also be time-consuming and may overlook relationships between different attributes. Furthermore, many prediction systems provide only the final classification without clearly indicating which factors contributed to the prediction. Therefore, a system that combines machine learning-based prediction with model evaluation and feature-importance analysis is highly useful for supporting early identification of students who may need academic intervention.

1.4 Proposed System

The proposed system is a machine learning-based student performance and dropout risk prediction framework.

The system accepts student academic, demographic, attendance, behavioural, social, and family-related data and performs the following high-level operations:

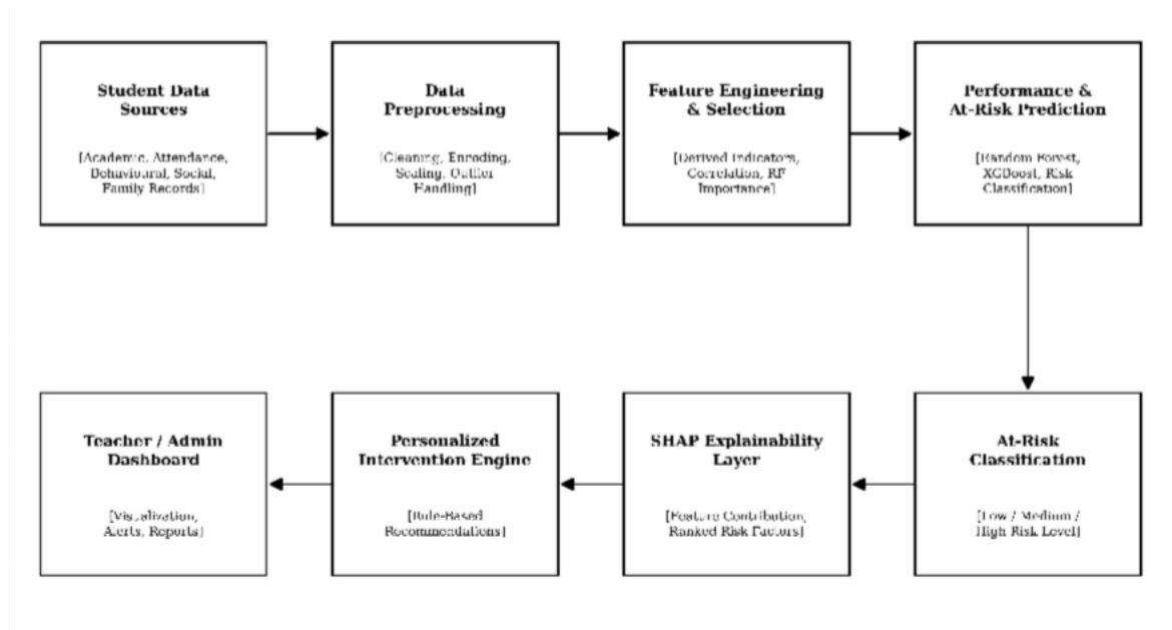


Figure 1.4: System Architecture

The system consists of the following major conceptual components:

1. Student Data Collection

2. Data Preprocessing
3. Feature Engineering and Selection
4. Performance Prediction
5. Risk Classification
6. Explainability Layer
7. Intervention Support
8. Teacher/Administrator Dashboard

The current implementation concentrates on **Data Preprocessing, Performance Prediction, Model Evaluation, and Feature Importance Analysis.**

Data Preprocessing

The module:

- cleans student data,
- handles missing values,
- encodes categorical features,
- prepares numerical features,
- divides the dataset for training and testing.

Performance Prediction

The prediction module:

- trains different machine learning models,
- compares their performance,
- performs cross-validation and tuning,
- predicts student performance as **Low, Average, or Good.**

Explainability

The explainability layer identifies the important factors contributing to a student's prediction and helps provide an understandable basis for the model's output.

Risk Identification

The system uses student performance and relevant factors to identify students who may require additional academic attention. The framework can later be extended to direct **Low / Medium / High dropout-risk prediction** using a suitable dropout-labelled dataset.

CHAPTER 2

LITERATURE SURVEY

2.1 Literature Review

The increasing use of educational data and machine learning techniques has created significant interest in predicting student academic performance and identifying students who may require early academic support. Existing research has explored the use of academic, demographic, behavioral, and social factors for predicting student outcomes. The following literature review examines important works related to student performance prediction, machine learning-based classification, feature analysis, and early identification of academically at-risk students.

2.1.1 Student Performance Datasets and Academic Prediction

Cortez and Silva (2008) studied student performance prediction using data collected from secondary schools and analysed academic, demographic, social, and behavioural factors. Their work demonstrated that machine learning techniques can be applied to predict student grades and identify factors associated with academic performance.

The UCI Student Performance Dataset has also been widely used as a benchmark for student performance prediction. It contains information related to students' academic results, family background, social activities, study habits, and other behavioural attributes. The dataset provides a suitable foundation for analysing relationships between student characteristics and academic outcomes.

2.1.2 Machine Learning-Based Student Performance Prediction

Several machine learning algorithms have been applied to student performance classification, including Logistic Regression, Decision Trees, Random Forest, Support Vector Machines, K-Nearest Neighbors, and Gradient Boosting. These approaches can identify complex patterns in student data and classify students according to their expected academic performance.

Ensemble learning methods such as Random Forest and Gradient Boosting are particularly useful because they can combine information from multiple features and capture non-linear relationships within the dataset. However, prediction performance can be affected by class imbalance, feature quality, and the selection of appropriate model parameters.

2.1.3 Feature Importance and Explainable Prediction

Academic prediction systems should not only provide a predicted outcome but should also help educators understand the factors influencing that prediction. Feature-importance

techniques can identify the attributes that contribute most significantly to model decisions. Explainable Artificial Intelligence techniques such as SHAP (SHapley Additive exPlanations) provide a way to interpret individual predictions by showing how different features contribute to the model output. Such approaches can improve the transparency of machine learning-based educational systems and make their predictions more useful for teachers and academic administrators.

2.1.4 Early Identification of At-Risk Students

Early identification of students who may experience poor academic outcomes is an important application of educational data mining. By analysing previous academic performance and related student characteristics, prediction systems can identify students who may require additional academic guidance.

However, many existing approaches focus primarily on predicting grades or final academic outcomes rather than providing an integrated early-warning framework. In addition, direct dropout prediction requires datasets containing reliable dropout-labelled records. Therefore, combining performance prediction, risk identification, and explainable machine learning can provide a stronger foundation for early academic intervention.

2.2 Comparison and Discussion

Study/System	Performance ML Prediction	Classification	Feature Analysis	Early Identification	Explainability
Cortez & Silva	✓	✓	Limited	Limited	✗
UCI Student Performance Dataset	✓	✓	✓	Limited	✗
Traditional ML Approaches	✓	✓	Limited	✓	Limited
SHAP-based Approaches	✓	✓	✓	✓	✓
Proposed System	✓	✓	✓	✓	✓

Table 2.2: Comparison of Existing Systems

Based on the existing literature, machine learning provides an effective approach for analysing student data and predicting academic outcomes. However, there is a need for systems that combine performance prediction, feature analysis, explainability, and early

identification within a single framework.

2.3 Conclusion

The literature review shows that machine learning techniques can be effectively used to analyse student-related data and predict academic performance. Existing studies have demonstrated the importance of academic, demographic, social, and behavioural factors in understanding student outcomes. Different classification algorithms have also been explored for predicting student performance, providing a strong foundation for developing predictive educational systems.

However, existing approaches often focus mainly on prediction and provide limited information about the factors influencing the predicted outcome. There is also a need for systems that can identify students requiring academic attention at an early stage rather than relying only on final examination results. Explainable machine learning techniques can further improve the usefulness and transparency of such prediction systems by showing the important factors contributing to a prediction.

Model comparison and proper evaluation are also important for selecting a reliable prediction approach. Techniques such as cross-validation and hyperparameter tuning can help assess model consistency and improve predictive performance. In addition, identifying important features can provide useful insights into the factors that have a stronger relationship with student outcomes.

Therefore, the proposed system integrates data preprocessing, machine learning-based performance prediction, model evaluation, feature-importance analysis, and early identification of students requiring academic support. The system provides a foundation for data-driven academic intervention and can help educators make more informed decisions regarding student support. It can be further extended to include direct dropout-risk prediction using a suitable labelled dataset, enabling a more comprehensive early-warning system for educational institutions.

CHAPTER 3

SYSTEM DESIGN

3.1 System Architecture

Fig. 3.1 shows the overall architecture of the proposed Machine Learning-Based Student Performance and Dropout Risk Prediction System. The system begins with student data sources containing academic, attendance, behavioural, social, and family-related information. The data is processed through the Data Preprocessing module for cleaning, encoding, and scaling, followed by Feature Engineering and Selection to identify relevant features.

The selected features are passed to the Performance and At-Risk Prediction module, where machine learning models such as Random Forest and XGBoost are used for prediction. The output is then processed by the At-Risk Classification module to identify the student's risk level. The SHAP Explainability Layer identifies the important factors influencing the prediction, while the Personalized Intervention Engine provides suitable recommendations. Finally, the Teacher/Administrator Dashboard presents visualizations, alerts, and reports for effective monitoring and decision-making.

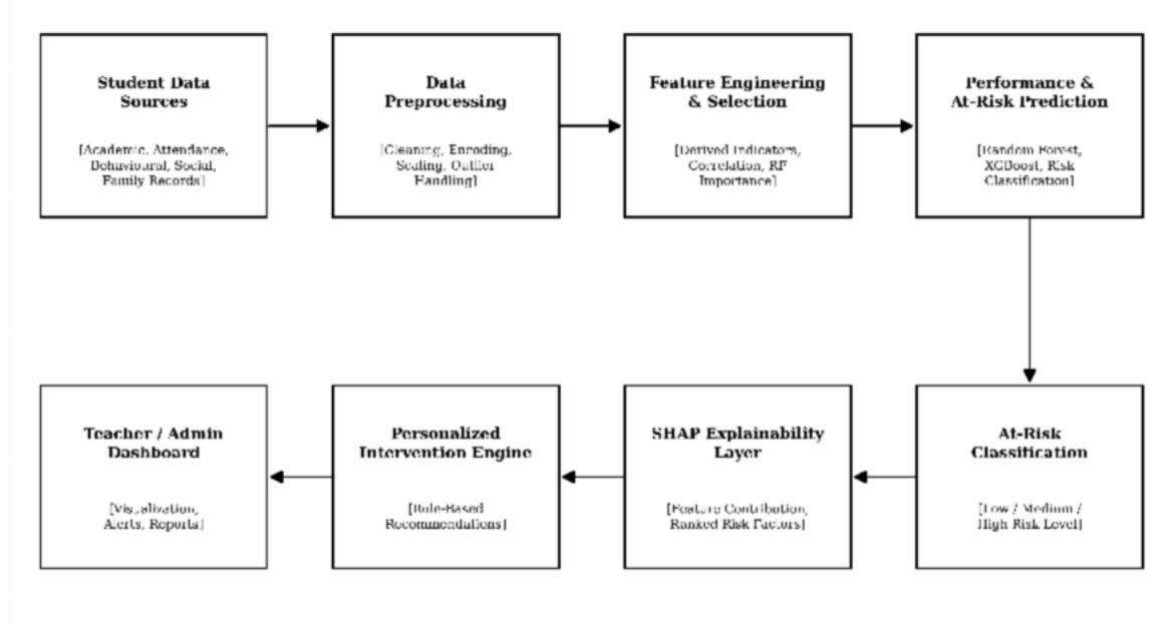


Figure 3.1: Overall architecture of the proposed machine learning-based student performance and dropout risk prediction system.

The first stage of the architecture is the Student Data Sources module. It acts as the input layer of the system and provides the information required for prediction. The data can

include academic records, attendance, behavioural factors, social information, and family-related records. These different attributes allow the system to consider multiple aspects of a student's background and academic situation rather than depending only on examination marks.

The collected information is then passed to the Data Preprocessing module. Raw student data may contain missing values, categorical attributes, inconsistent values, or features with different numerical scales. Therefore, preprocessing is performed before applying machine learning algorithms. The module includes operations such as data cleaning, categorical encoding, scaling, and required data handling. The resulting processed dataset provides a consistent input for the subsequent stages of the architecture.

After preprocessing, the data moves to the Feature Engineering and Selection module. This stage focuses on identifying the features that are useful for predicting student performance. Derived indicators can be obtained from existing attributes, while correlation analysis and model-based feature importance can be used to understand the contribution of different features. Feature selection helps the prediction stage concentrate on meaningful student characteristics and reduces the influence of unnecessary information.

The selected features are then supplied to the Performance and At-Risk Prediction module. This is the main machine learning stage of the architecture. Machine learning algorithms are trained using the prepared student dataset to learn relationships between input features and student outcomes. Models such as Random Forest and XGBoost can be used within the proposed architecture for prediction. The prediction stage produces the expected performance outcome and information that can subsequently be used for identifying students who may require attention.

The prediction output is passed to the At-Risk Classification module. This stage converts the model output into an understandable risk category. The proposed architecture represents risk using Low, Medium, and High Risk Levels. Such classification makes the prediction easier for educators to interpret and helps them prioritize students who may require greater academic support.

The next stage is the SHAP Explainability Layer. A prediction alone does not explain why a student has received a particular result. Therefore, this layer analyses the contribution of individual features to the model's prediction. SHAP-based feature contribution can identify and rank the factors that have the greatest influence on a particular student's prediction. This improves the transparency of the system and allows educators to understand the reasoning behind the machine learning output.

The information generated by the prediction and explainability stages is then provided to the Personalized Intervention Engine. This component uses the identified risk factors and prediction results to support rule-based recommendations. For example, students requiring academic attention can be identified for additional monitoring or appropriate academic guidance. The purpose of this stage is to convert prediction results into useful intervention information rather than limiting the system to classification alone.

Finally, the processed results are presented through the Teacher/Administrator Dashboard. The dashboard acts as the output and monitoring layer of the architecture. It can display student performance information, risk levels, visualizations, alerts, and reports. This provides teachers and administrators with a centralized view of the students who may require attention and supports data-driven academic decision-making.

Overall, the architecture establishes a complete flow from student data collection → preprocessing → feature selection → machine learning prediction → risk classification → explainability → intervention → dashboard visualization. The current implementation primarily establishes the data processing and student performance prediction pipeline, while the architecture provides a foundation for extending the system towards direct dropout-risk prediction when suitable dropout-labelled data is incorporated.

3.2 System Requirements

This section describes the software requirements, hardware requirements, dataset requirements, and deployment requirements needed for the proposed Machine Learning-Based Student Performance and Dropout Risk Prediction System.

3.2.1 Software Requirements

Component	Requirement
Operating System	Windows/Linux
Programming Language	Python
Development Environment	Jupyter Notebook / Google Colab
ML Framework	Scikit-learn
Visualization	Matplotlib, Seaborn

Explainable AI	SHAP
Data Storage	CSV / Pandas DataFrame
Input Format	CSV
Output Format	Prediction Report / DataFrame
Prediction Output	Low / Average / Good
Prediction Probability	Class Probability Scores

Table 3.2.1: Software Requirements

SHAP fits into the explainability module since it helps identify how individual student features contribute to the model's prediction. It enables us to understand the influence of attributes such as previous grades, age, absences, and past failures on the predicted academic performance, making the machine learning results more transparent and interpretable.

3.2.2 Hardware Requirements

Proper provisioning of hardware resources is required to support the smooth development, testing, and execution of the student performance prediction system. The proposed machine learning pipeline involves data preprocessing, feature transformation, model training, cross-validation, hyperparameter tuning, and prediction, which primarily require sufficient CPU processing power and system memory. Since the implemented models are based on conventional machine learning algorithms rather than computationally intensive deep learning models, a dedicated GPU is not mandatory. The table below provides the minimum and recommended hardware requirements for running the system. The minimum configuration is sufficient for basic development, preprocessing, model training, debugging, and testing, while the recommended configuration provides additional processing capability and memory for repeated model training, cross-validation, and larger datasets.

Component	Minimum Configuration	Recommended Configuration
Processor (CPU)	Intel Core i3 / Equivalent Multi-Core CPU	Intel Core i5 or above
System Memory (RAM)	4 GB	8 GB or above
Graphics (GPU)	Not Required	Optional NVIDIA GPU
Storage (SSD)	256 GB SSD	512 GB or above SSD

Table 3.2.2: Hardware Requirements

Dataset Requirements:

Student Performance Dataset

Dataset includes:

- Student Performance Dataset — academic and demographic information
- 395 student records
- 33 original features
- 32 input features after target removal
- 58 processed features after encoding and transformation
- 316 training records
- 79 testing records
- Low, Average, Good — performance classes

The Student Performance Dataset is used to develop and evaluate the machine learning-based student performance prediction system. It contains academic, demographic, social, and behavioural attributes of students that can be used to identify factors associated with academic performance. During preprocessing, missing and inconsistent values are handled, categorical attributes are encoded, and the input features are transformed into a suitable format for machine learning. The original four-class target consisting of Low, Average, Good, and Excellent was converted into three classes because the Excellent category

contained only 19 students and resulted in severe class imbalance. The resulting classes are Low, Average, and Good, which are used for training and evaluating the prediction models. The dataset is divided into training and testing subsets, with 316 records used for training and 79 records for testing. Multiple machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, KNN, SVM, and Gradient Boosting, are evaluated using Accuracy, Precision, Recall, and F1-score. The dataset currently supports academic performance prediction; a separate labelled dropout dataset will be required to develop and validate an actual dropout-risk prediction model.

3.3 Activity Diagram

The activity diagram illustrates the step-by-step flow of the proposed system, starting with loading the student dataset and performing data preprocessing such as cleaning, encoding, and scaling. The processed data then moves through feature selection and engineering before being split into training and testing sets. Multiple machine learning models are trained and evaluated using cross-validation and hyperparameter tuning to identify the best-performing model based on accuracy, precision, recall, and F1-score. The selected model generates predictions classifying students into Low, Average, or Good performance categories, which are then explained using SHAP to highlight the key contributing factors. Finally, the results and explanations are stored and displayed as output for educators, as illustrated in the diagram below.

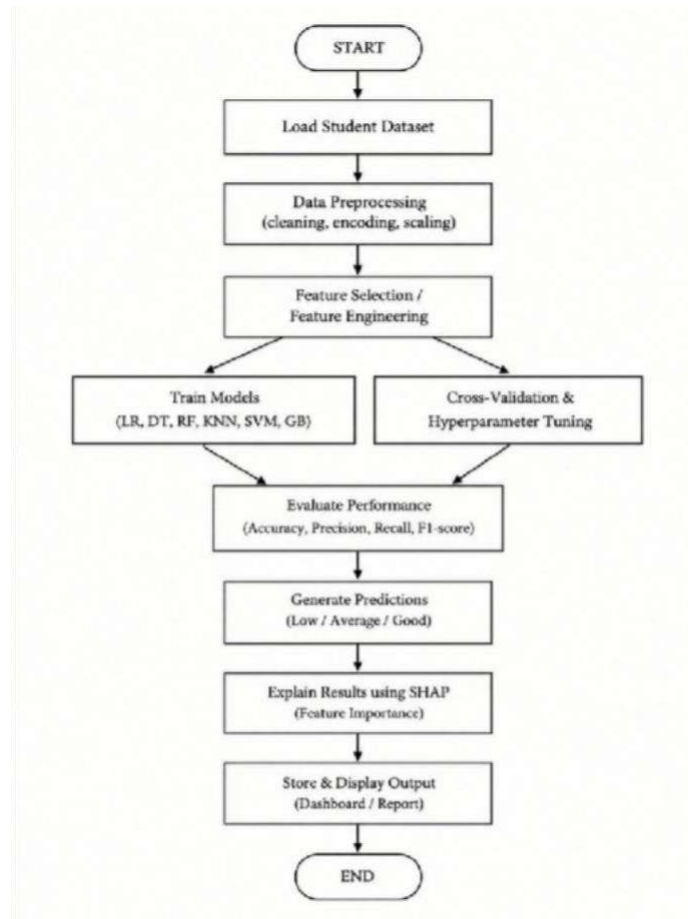


Figure 3.3: Activity Diagram

3.4 Data Flow Diagram

The Data Flow Diagram illustrates the complete flow of the student performance prediction system, starting with the student dataset and moving through data collection and preprocessing. The processed data is passed to feature selection and model training, while cross-validation and hyperparameter tuning are used to optimize and evaluate the models. The processed features are also analyzed using SHAP to identify important factors influencing predictions. Finally, the trained model generates the student's predicted performance category as Low, Average, or Good along with probability scores, which are stored and presented as the final output.

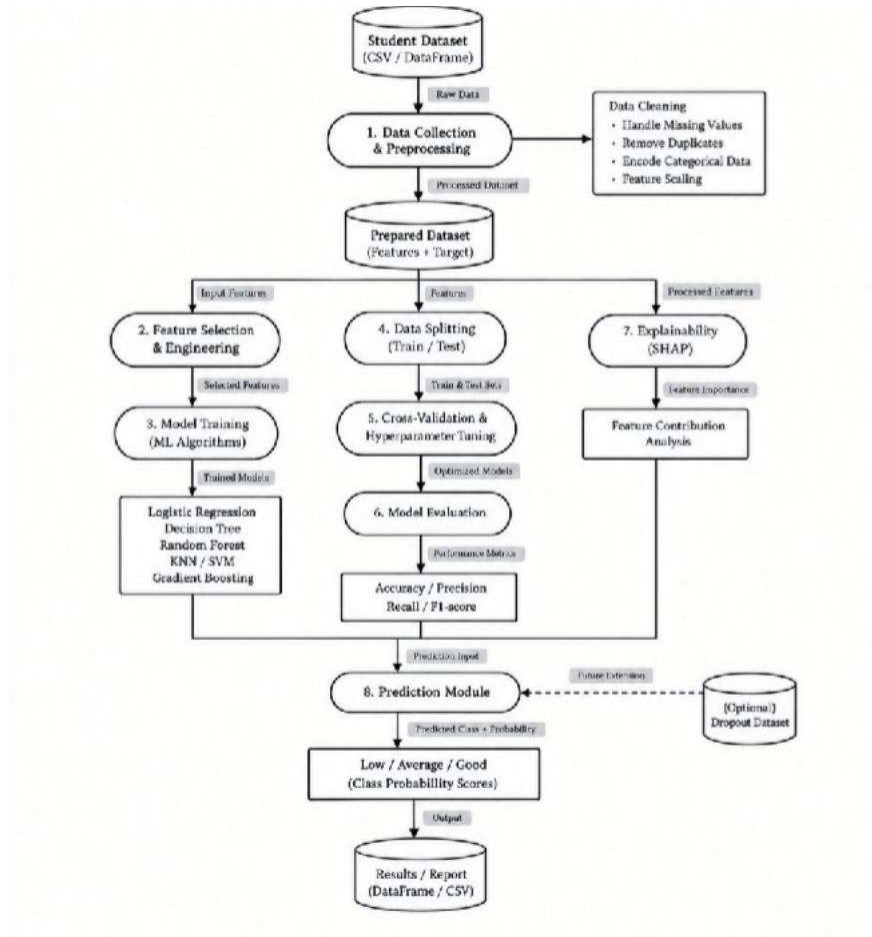


Figure 3.4: Data Flow Diagram

CHAPTER 4

PROJECT DESCRIPTION

4.1 Methodologies

The proposed approach follows a modular machine learning-based prediction approach. Rather than treating student performance prediction as a single classification task, the system divides the process into several stages including data preprocessing, feature engineering, model training, evaluation, prediction, and explainability.

4.1.1 Machine Learning-Based Performance Prediction

The prediction module handles the complete process of training and evaluating machine learning models, from preparing the student data to generating the final performance

category. Multiple classification algorithms are trained and compared to determine the most suitable model for predicting student academic performance. This approach enables systematic evaluation of different models and supports reliable identification of students who may require academic attention.

4.1.2 Data Preprocessing and Feature Engineering

Student data is processed before being provided to the prediction models. The preprocessing stage handles missing values, removes duplicate or inconsistent records, encodes categorical attributes, and applies suitable feature transformations and scaling.

The processing order is as follows:

Stage 1: Data Cleaning

The raw student dataset is cleaned by handling missing values, removing duplicate records, and correcting inconsistent data. This ensures that the input provided to the machine learning models is reliable and suitable for analysis.

Stage 2: Feature Transformation

Categorical attributes are converted into numerical representations and the required numerical features are scaled. Feature engineering and selection are then performed to prepare the most relevant attributes for model training.

Stage 3: Model Training and Evaluation

The processed dataset is divided into training and testing sets. Models such as Logistic Regression, Decision Tree, Random Forest, KNN, SVM, and Gradient Boosting are trained and evaluated using Accuracy, Precision, Recall, and F1-score. Cross-validation and hyperparameter tuning are also applied to improve model performance.

Stage 4: Explainable Prediction

SHAP is used to analyze the contribution of individual features to the model's predictions. This helps identify important factors influencing student performance and makes the prediction process more interpretable. The final system generates the predicted category as **Low, Average, or Good**, along with the corresponding prediction probabilities.

4.2 Modules

4.2.1 Data Collection and Preprocessing Module

The Data Collection and Preprocessing Module is responsible for receiving the student

performance dataset and preparing it for machine learning.

Functions

- Dataset loading
- Missing value handling
- Duplicate removal
- Inconsistent data handling
- Categorical data encoding
- Numerical feature transformation
- Feature scaling
- Train-test data preparation

The module processes the raw student records and converts them into a structured format suitable for machine learning models. The dataset contains academic, demographic, social, and behavioural attributes that are cleaned and transformed before further processing.

Output

Processed Student Dataset



Cleaned Features + Target

4.2.2 Feature Selection and Engineering Module

This module prepares the most relevant features for predicting student academic performance. After preprocessing, categorical attributes are converted into numerical representations and the resulting features are organized for model training.

A. Feature Transformation

Categorical variables are encoded into numerical values so that they can be processed by machine learning algorithms. Numerical attributes are also transformed and scaled where required.

B. Feature Selection

The module identifies relevant student attributes that contribute to academic performance. Features such as previous grades, age, absences, past failures, and behavioural attributes are considered during the modelling process.

The processed dataset contains **58 features** after encoding and transformation. This stage reduces unnecessary complexity and provides suitable input features for the prediction models.

4.2.3 Machine Learning Model Training

The Machine Learning Model Training Module is responsible for training multiple classification algorithms using the prepared student dataset.

The implemented models include:

- Logistic Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbors
- Support Vector Machine
- Gradient Boosting

The dataset is divided into **316 training records and 79 testing records**. Each model is trained using the training data and subsequently evaluated on unseen test data.

The target variable is classified into three performance categories:

Low → Average → Good

The original four-class classification included an **Excellent** category, but it contained only 19 students and resulted in severe class imbalance. Therefore, the three-class classification was adopted for more reliable model evaluation.

4.2.4 Cross-Validation and Hyperparameter Tuning

The Cross-Validation and Hyperparameter Tuning Module is used to improve the reliability and performance of the trained models.

A. Stratified Cross-Validation

Five-fold stratified cross-validation is applied to maintain the distribution of the performance classes across different folds. This provides a more reliable estimate of model performance than relying only on a single train-test split.

B. Hyperparameter Tuning

Grid Search is applied to selected machine learning models to identify suitable hyperparameter combinations. Random Forest, Decision Tree, and Gradient Boosting are tuned using cross-validation.

The best cross-validation F1-scores obtained were **0.8700 for Random Forest, 0.8803 for Decision Tree, and 0.8891 for Gradient Boosting**.

4.2.5 Model Evaluation Module

The Model Evaluation Module compares the performance of the trained classification models using standard evaluation metrics.

The evaluation metrics include:

- Accuracy
- Precision
- Recall
- F1-score

The models are evaluated using the test dataset to determine their ability to correctly classify student performance. The tuned **Gradient Boosting model** was selected as the final model, achieving **87.34% Accuracy, 87.95% Precision, 87.34% Recall, and 87.42% F1-score**. This evaluation provides a basis for selecting the most suitable model for the final prediction module.

4.2.6 Student Performance Prediction Module

The Student Performance Prediction Module uses the selected machine learning model to generate predictions for new student records.

The input student attributes are processed using the same preprocessing and feature transformation steps applied during model training. The trained Gradient Boosting model then predicts the student's academic performance category.

The output consists of:

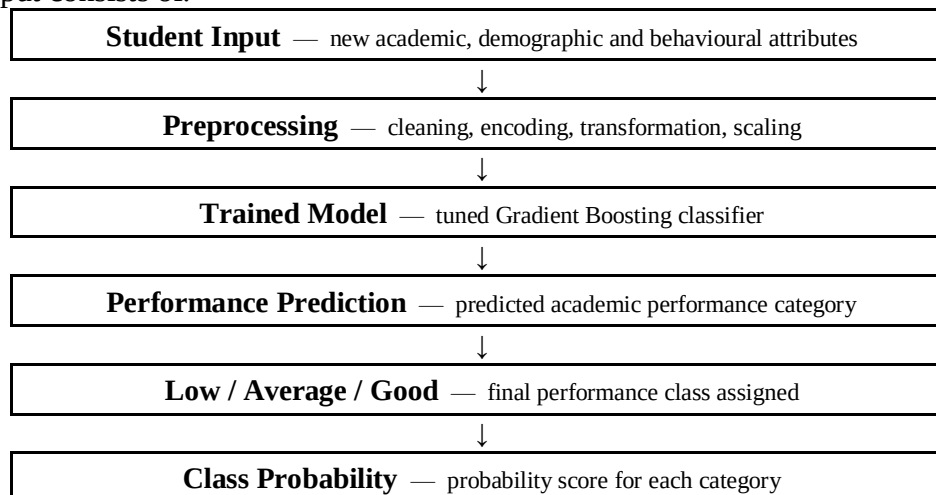


Figure 4.1: Student Performance Prediction Output Flow

For example, the system can generate probability scores for each performance category and identify the most probable category as the final prediction.

4.2.7 Explainable AI Module

The Explainable AI Module is used to understand the factors influencing the predictions generated by the machine learning model. **SHAP (SHapley Additive exPlanations)** is used to analyze feature contributions and improve the interpretability of the prediction results.

The module identifies the features that have the greatest influence on student performance. In the implemented model, **G2** is the most influential feature, followed by attributes such as **G1, Age, Going Out Frequency, alcohol consumption, parental occupation, Absences, and Past Failures.**

This module makes the prediction process more transparent by showing **why a particular performance category was predicted**, rather than providing only the final classification.

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

5.1 Implementation and Results

This document details the implementation of the **AI-Based Student Academic Performance and Risk Prediction System**. The developed system includes student data management, academic performance prediction, risk analysis, model comparison, prediction history, and explainability features. The implementation provides an interactive interface through which the administrator can view student information, generate predictions, analyse model performance, and understand the factors influencing individual predictions.

5.1.1 Data Pre-processing

Conversion of the raw student dataset into a clean and structured format suitable for machine learning forms the first step of the implementation pipeline.

Operations

The system performs the following preprocessing operations:

Missing Value Handling: Missing values are identified and appropriately handled to maintain consistency in the dataset.

Duplicate Removal: Duplicate and inconsistent records are removed to reduce unnecessary bias during model training.

Categorical Encoding: Categorical student attributes are converted into numerical representations suitable for machine learning algorithms.

Feature Transformation: Input attributes are transformed into a consistent numerical format before training and prediction.

Feature Scaling: Numerical features are scaled where required to provide suitable input for algorithms such as KNN, SVM, and Logistic Regression.

After preprocessing and encoding, the original **33 features are transformed into 58 processed features** for model training.

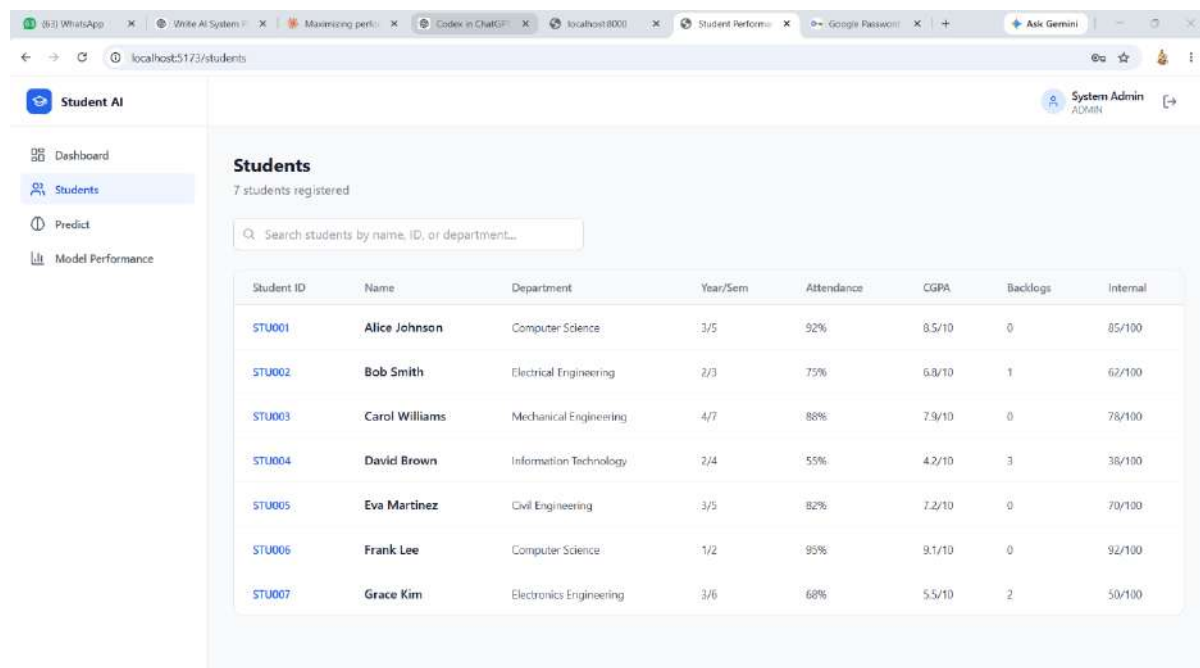
5.1.2 Dataset Preparation and Student Management Interface

The Student Performance Dataset contains **395 student records** with academic, demographic, social, and behavioural attributes. The processed dataset is divided into training and testing subsets for model development and evaluation.

Dataset Component	Records / Features
Total Records	395
Original Features	33
Processed Features	58
Training Records	316
Testing Records	79

Table 5.1.2: Dataset Preparation

The developed system also provides a **Student Management Interface** for viewing and accessing student information. As shown in the implemented interface, the administrator can view details such as **Student ID, Name, Department, Year/Semester, Attendance, CGPA, Backlogs, and Internal Marks**. A search facility is also provided to locate students using their name, ID, or department.



The screenshot displays a web application interface for 'Student AI'. The left sidebar contains navigation links: Dashboard, Students (active), Predict, and Model Performance. The main content area is titled 'Students' and shows '7 students registered'. Below this is a search bar with the placeholder text 'Search students by name, ID, or department...'. A table lists the following student data:

Student ID	Name	Department	Year/Sem	Attendance	CGPA	Backlogs	Internal
STU001	Alice Johnson	Computer Science	3/5	92%	8.5/10	0	85/100
STU002	Bob Smith	Electrical Engineering	2/3	75%	6.8/10	1	62/100
STU003	Carol Williams	Mechanical Engineering	4/7	88%	7.9/10	0	78/100
STU004	David Brown	Information Technology	2/4	55%	4.2/10	3	38/100
STU005	Eva Martinez	Civil Engineering	3/5	82%	7.2/10	0	70/100
STU006	Frank Lee	Computer Science	1/2	95%	9.1/10	0	92/100
STU007	Grace Kim	Electronics Engineering	3/6	68%	5.5/10	2	50/100

Figure 5.1.2: Student Management Interface

The student management module acts as the primary input and monitoring component of the system. It allows student records to be organised and reviewed before generating predictions. The displayed information provides a structured overview of academic indicators that can be used by the prediction system for further performance and risk analysis.

5.1.3 Machine Learning Model Results

The system evaluates multiple machine learning classification algorithms to identify the most suitable model for student academic performance prediction.

The implemented models include:

- Logistic Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbors
- Support Vector Machine
- Gradient Boosting

Model Performance

Model	Accuracy	F1-Score
Decision Tree	87.34%	87.31%
Logistic Regression	86.08%	86.19%
Random Forest	86.08%	86.16%
Gradient Boosting	84.81%	84.89%
SVM	78.48%	78.36%
KNN	70.89%	71.07%

Table 5.1.3: Machine Learning Model Performance

The initial results show that the Decision Tree achieved the highest test accuracy among the initially evaluated untuned models, while the other algorithms demonstrated different levels of performance across the student performance categories.

5.1.4 Cross-Validation Results

To assess the consistency of the models across different subsets of the dataset, 5-fold stratified cross-validation was performed.

Model	Mean F1-Score	Standard Deviation
Random Forest	0.8534	0.0653
Gradient Boosting	0.8509	0.0297
Decision Tree	0.8342	0.0237

Model	Mean F1-Score	Standard Deviation
Logistic Regression	0.8312	0.0486
SVM	0.7979	0.0306
KNN	0.6637	0.0394

Table 5.1.4: 5-Fold Stratified Cross-Validation Results

The cross-validation results show that Random Forest achieved the highest mean F1-score among the initially evaluated models, while Gradient Boosting demonstrated competitive performance with comparatively lower variation across folds.

5.1.5 Hyperparameter Tuning and Final Model Evaluation

Grid Search was applied to selected machine learning models to identify suitable hyperparameter combinations and improve their predictive performance.

Model	Best Cross-Validation F1-Score
Random Forest	0.8700
Decision Tree	0.8803
Gradient Boosting	0.8891

Table 5.1.5: Hyperparameter Tuning Results

The tuned Gradient Boosting model achieved the highest cross-validation F1-score of 0.8891 and was selected as the final academic performance prediction model.

The selected model was then evaluated on the testing dataset.

Metric	Result
Accuracy	87.34%
Precision	87.95%
Recall	87.34%
F1-Score	87.42%

Table 5.1.6: Final Model Evaluation Results

The final model achieves an accuracy of 87.34% with an F1-score of 87.42%, demonstrating its ability to classify students into the defined performance categories.

5.1.6 Dashboard and Risk Analysis

The developed system includes an **administrative dashboard** that provides a consolidated overview of student performance and prediction-related information.

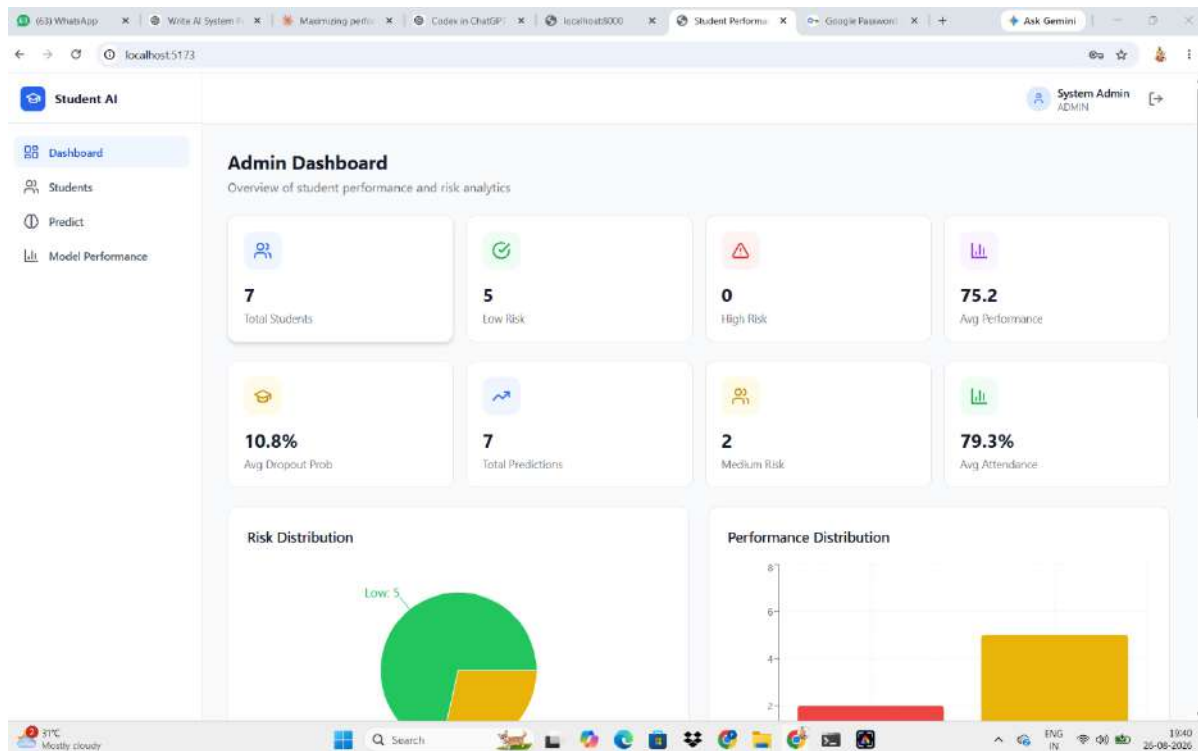


Figure 5.1.6: Administrative Dashboard and Risk Analysis

The dashboard displays important system-level indicators, including the total number of registered students, low-risk students, medium-risk students, high-risk students, average academic performance, average dropout probability, and total predictions generated.

In the implemented demonstration shown in the interface, the dashboard displays 7 registered students, with 5 classified under Low Risk and 2 under Medium Risk, while no student is currently displayed under High Risk. The dashboard also presents an average performance value of 75.2 and an average attendance of 79.3% for the displayed student records.

The visual analytics section further includes a Risk Distribution chart and a Performance Distribution chart. These charts provide a quick graphical summary of the prediction outcomes and help administrators identify the overall distribution of students across

different categories.

This dashboard improves the usability of the system by presenting prediction results and student-level trends in a single interface rather than requiring manual examination of individual records.

5.1.7 Recent Predictions and Prediction History

The system stores generated predictions and displays them through a Recent Predictions Interface.

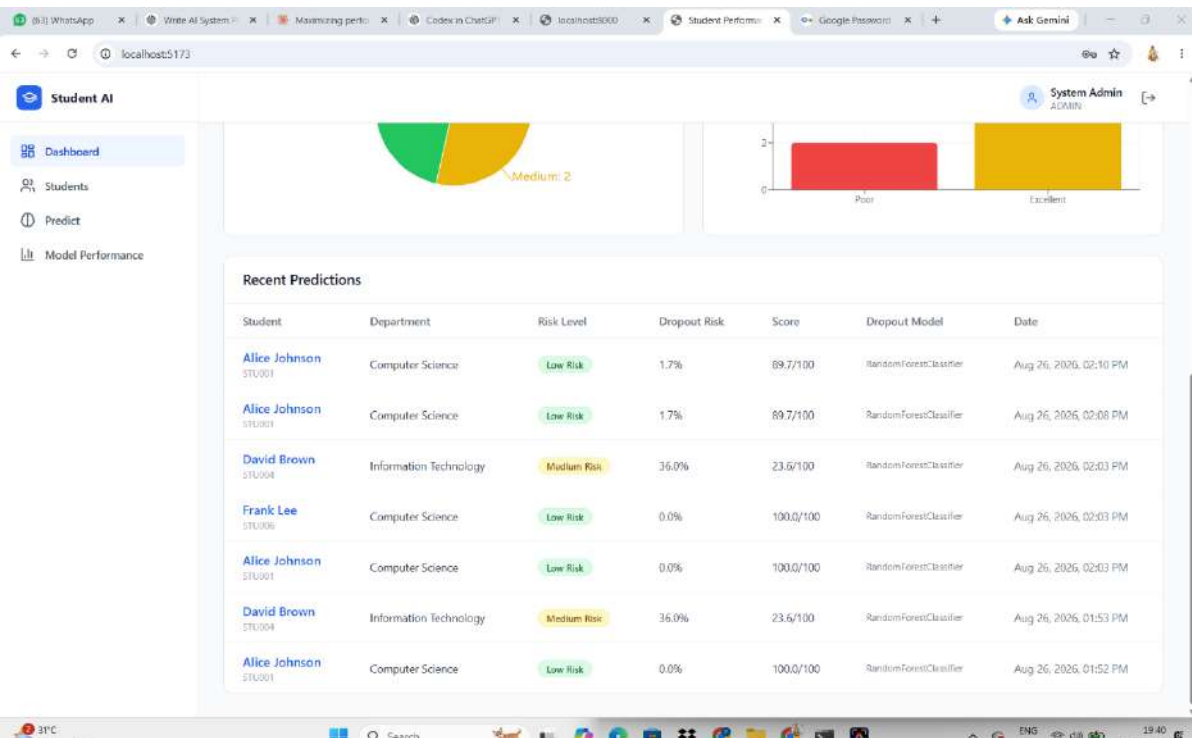


Figure 5.1.7: Recent Student Predictions

The prediction history interface displays previously generated results in a structured tabular format. The displayed fields include Student Name, Student ID, Department, Risk Level, Dropout Risk, Performance Score, Prediction Model, and Date/Time of Prediction. The screenshot demonstrates how individual prediction records can be monitored after model execution. For example, the interface distinguishes students using risk labels such as Low Risk and Medium Risk, while also displaying the corresponding numerical dropout-risk percentage and academic score.

This functionality is useful for maintaining a historical record of predictions and enables

administrators to compare student results over time. It also supports the traceability of the system by recording the model associated with each prediction and the time at which the result was generated.

Important: Since your actual dataset currently does not contain a properly labelled dropout target, keep the discussion careful in the final report. You can describe this interface as the implemented risk-monitoring/output component, but don't claim that the displayed dropout percentages are experimentally validated dropout-model results unless you have separately trained and validated that model.

5.1.8 Model Performance Comparison

The implementation includes a dedicated **Model Performance Interface** for comparing the machine learning algorithms used within the system.

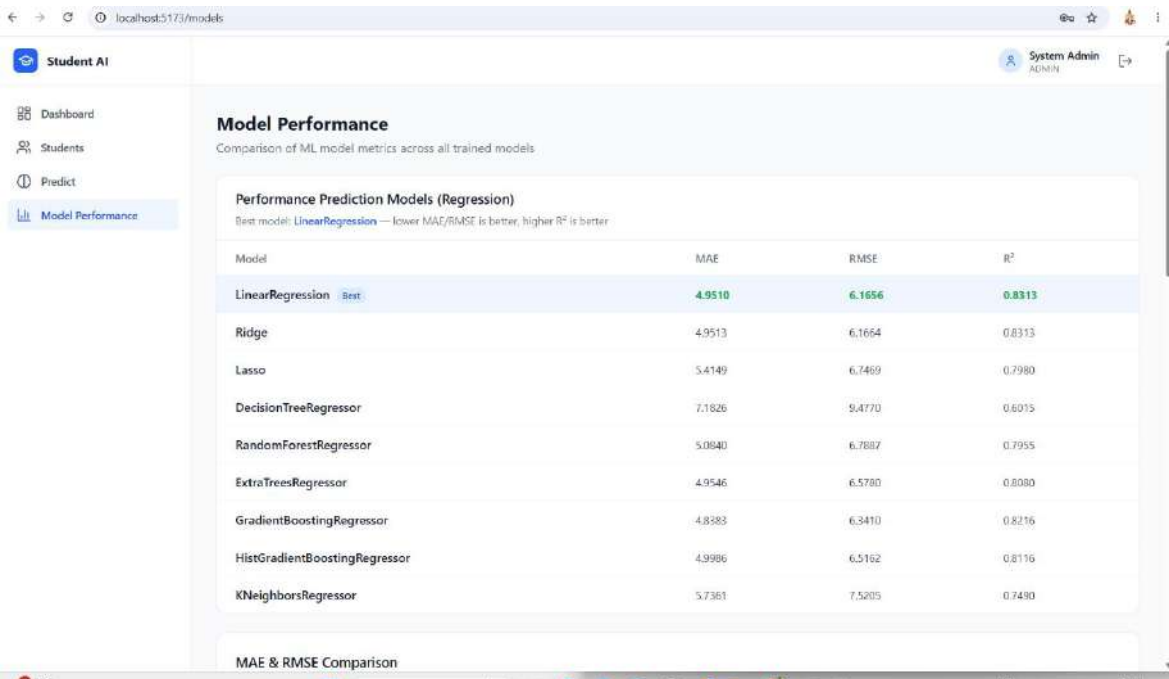


Figure 5.1.8: Machine Learning Model Performance Comparison

The interface presents the performance of multiple classification models in tabular and graphical form. The models displayed include algorithms such as Logistic Regression, Decision Tree, Random Forest, Extra Trees, Gradient Boosting, AdaBoost, K-Nearest Neighbors, Support Vector Classifier, and Histogram Gradient Boosting, depending on the

implemented comparison configuration.

The graphical comparison allows the administrator to observe differences between evaluation metrics such as Accuracy, F1-Score, and ROC-AUC. The visual representation makes it easier to compare models than analysing numerical values alone.

The model comparison module is particularly useful during the development phase because it supports the selection of an appropriate algorithm based on more than one evaluation metric. Instead of selecting a model solely according to accuracy, the system can compare predictive performance across different measures.

5.1.9 Prediction Explainability Using SHAP

To improve the transparency of the prediction process, the system incorporates a SHAP-based Prediction Explanation Module.

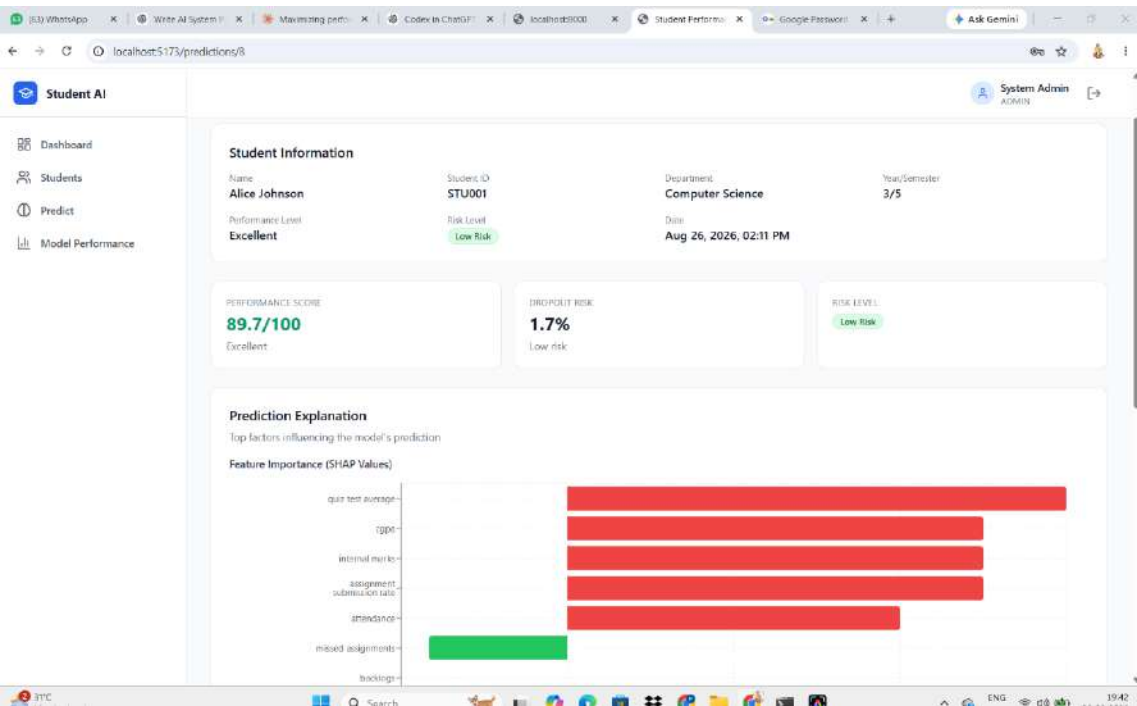


Figure 5.1.9: SHAP-Based Prediction Explanation

The explainability interface displays the features that have the greatest influence on an individual student's prediction. In the implemented demonstration, factors such as Quiz Test Average, CGPA, Internal Marks, Assignment Submission Rate, Attendance, Missed Assignments, and Backlogs are visualised according to their contribution to the generated

result.

The feature contributions are displayed through a horizontal bar chart, allowing positive and negative influences to be distinguished visually. The interface also separately identifies Positive Factors and Negative Factors affecting the prediction.

For the example shown in the system, higher values of academic indicators such as Quiz Test Average, CGPA, and Internal Marks are displayed as major positive contributors, while Missed Assignments contribute negatively to the prediction. Other factors, including attendance and assignment submission behaviour, also influence the generated result.

This explainability component is important because the system does not provide only a final performance category. It also gives the administrator an indication of why a particular prediction was produced. Such information can help educators identify the academic and behavioural factors associated with a student's predicted outcome and potentially take appropriate intervention measures.

Revised 5.2 Discussion

Unlike conventional academic evaluation systems that mainly depend on manual analysis of marks and periodic assessments, the proposed system integrates student data management, machine learning-based prediction, model comparison, risk visualisation, prediction history, and explainability within a single interface.

The implementation screenshots demonstrate the practical operation of the system. The Student Management Interface provides structured access to student records and academic information, while the administrative dashboard presents overall statistics such as student count, risk distribution, average performance, attendance, and prediction summaries. The Recent Predictions module further provides a record of generated results, including the associated risk level, score, prediction model, and prediction date.

The system also supports the comparison of multiple machine learning algorithms through the Model Performance Interface. This provides a visual and tabular representation of different evaluation metrics and assists in analysing the relative performance of the implemented models. Based on the reported experimental results, the tuned Gradient Boosting model achieved 87.34% accuracy and an F1-score of 87.42% on the test dataset.

An additional strength of the implementation is the integration of SHAP-based explainability. The Prediction Explanation Interface identifies the features contributing to an individual result and separates positive and negative influences. In the demonstrated

interface, academic factors such as quiz performance, CGPA, internal marks, assignment submission, and attendance are displayed as important contributors, while missed assignments and backlogs can negatively affect the prediction.

However, the current dataset does not contain a labelled dropout target. Therefore, the experimentally validated results presented in this work relate specifically to student academic performance prediction. The risk and dropout-related interface components provide a framework for early identification and future risk monitoring, but actual dropout prediction performance should only be claimed after training and validating the system using an appropriate dataset containing verified dropout labels.

My recommendation

This looks much better if you use the screenshots in this order:

1. **Students page** → Figure 5.1.2
2. **Dashboard** → Figure 5.1.6
3. **Recent Predictions** → Figure 5.1.7
4. **Model Performance** → Figure 5.1.8
5. **SHAP Explanation** → Figure 5.1.9

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

The proposed system presents a machine learning-based framework for analysing student academic performance and supporting the early identification of students who may require academic attention. It addresses the limitations of conventional academic evaluation systems by analysing multiple student attributes and applying machine learning techniques to generate data-driven performance predictions.

The developed system focuses on important stages including data preprocessing, feature engineering, model training, performance evaluation, prediction, and explainability.

The Data Preprocessing Module prepares the student dataset by handling missing values, removing duplicate and inconsistent records, encoding categorical attributes, and transforming the data into a suitable format for machine learning. Multiple classification

models including Logistic Regression, Decision Tree, Random Forest, KNN, SVM, and Gradient Boosting are evaluated to identify the most suitable prediction approach.

The final tuned Gradient Boosting model achieved 87.34% accuracy, 87.95% precision, 87.34% recall, and 87.42% F1-score on the test dataset. The integration of SHAP-based explainability further helps in understanding the factors influencing the model's predictions, with previous academic performance, particularly G2, showing the strongest contribution. Therefore, the proposed framework moves beyond simple academic score analysis towards an explainable and data-driven early identification system that can provide performance predictions, probability scores, and feature-based explanations to support timely academic intervention.

6.2 Future Work

Several improvements can be incorporated into future versions of the system to extend its capabilities and improve its usefulness for educational institutions. The most important extension is the development of a dedicated **dropout-risk prediction module** using a suitable labelled dataset containing historical dropout information. Such a dataset would allow machine learning models to directly classify students according to their likelihood of dropping out and enable proper evaluation of dropout prediction performance.

The system can further be improved by incorporating additional real-time student information such as attendance, assignment submission, examination results, learning management system activity, and semester-wise academic progress. Continuous collection of these attributes could enable the system to monitor changes in student behaviour and provide earlier warnings instead of relying only on static records.

Another possible extension is the development of a teacher and administrator dashboard through which predicted performance, risk levels, important contributing factors, and student progress can be visualized. The explainability module can also be enhanced using SHAP-based individual student explanations so that educators can understand the reasons behind each prediction.

The prediction framework can additionally be evaluated using larger and more diverse datasets from different institutions to improve generalization. Future versions may also explore advanced ensemble and deep learning approaches where sufficient data is available. Finally, a recommendation and intervention module can be incorporated to suggest suitable academic support strategies for identified at-risk students, transforming the system from a prediction tool into a more complete early-warning and student-support framework.

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7

Explainable AI-Based Student Performance and Dropout Risk Prediction for Early Identification of At-Risk Students

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Abstract—Now, huge amounts of educational data are generated by the learning management system, attendance, and continuous assessment process, thereby enabling the identification of students who are likely to encounter difficulties or dropout well before performance declines. Conventional monitoring practices, which are largely based on performance assessment and teacher's observations, are inadequate in addressing the multiple variables and complex interactions that determine a student's progress. Machine Learning (ML) has proven to be a reliable solution for recognizing patterns within students' records, but these patterns are sometimes difficult to interpret, thus rendering them ineffective. This paper proposes an Explainable AI-based framework for predicting student performance and identifying students at academic risk while providing transparency by interpreting the reasons behind the predictions using SHAP (SHapley Additive exPlanations). The framework uses academic records, attendance, behavior, family, and social features extracted from the UCI Student Performance dataset to predict student performance and academic risk. It is designed using the final grade (G3) as a performance metric and developing an alternative indicator for the absence of a dropout label. The proposed methodology involves data preprocessing, feature engineering and selection, performance and at-risk prediction using Random Forest and XGBoost, SHAP-based model interpretation, and a rule-based personalized intervention module, which transforms identified risk factors into interpretable insights for the end users. By prioritizing prediction transparency and providing relevant recommendations for intervention, the framework is expected to become more than just a prediction model but a trusted advisor to help teachers intervene with at-risk students before it is too late.

Keywords—Predicting student performance; predicting dropout risk; machine learning; explainable AI; SHAP; educational data mining; early intervention; student analytics.

I. INTRODUCTION

The rapid adoption of learning management systems has led to the generation of a large amount of student-related data. Currently, educational institutions collect student records, which contain information on their academic performance, attendance, behavior, study habits, family details, and level of engagement in learning activities. This volume and variety of data can be leveraged using Artificial Intelligence (AI) and Machine Learning (ML) to monitor academic performance and intervene early with students who are at academic risk or whose performance is below average.

A student's performance is a function of multiple variables, some of which include previous grades and scores, number of hours spent studying, school attendance record, family's

academic encouragement, educational background, and social circle among others. Traditional performance monitoring practices are largely based on performance assessment and teacher's observation, which are primarily subjective measures.

These approaches are mostly ineffective in addressing the complex interactions and many variables that contribute to a student's performance. With the ability to identify patterns within students' records, Machine Learning (ML) has proven to be a reliable solution for predicting performance and identifying students at academic risk. However, these predictions can be difficult to interpret, thereby rendering them ineffective. Explainable Artificial Intelligence (XAI) addresses this issue by providing a human understandable explanation of the underlying factors that lead to a certain prediction. In this context, SHAP can be used to determine the contribution of each feature to a given prediction, thus explaining what causes a student's performance to decline or improve.

This project involves developing an innovative approach comprising of performance prediction, at-risk prediction, explainable AI, and personalized recommendation modules into a single framework that can be trusted to support early intervention decisions. It aims to combine attendance, behavior, socio-economic, and academic data to provide explanations for performance predictions as well as recommendations for improving performance and addressing the causes of performance decline. The UCI Student Performance dataset will be used as the primary source of student records on which the framework will be trained and evaluated. The dataset comprises of 395 instances with 33 attributes that contain information on a student's demographics, family, schools, social life, lifestyle, and academic performance. The objective will be to predict final grade (G3), which is the final term Mathematics grade, using other attributes. A limitation of this dataset is that it does not include a class label for students who dropped out of school or were withdrawn by their parents. Instead, an alternative approach will be used to determine whether a student is at academic risk or not based on their number of failures prior to the final term, attendance record, and other attributes that indicate such risks. The proposed solution involves developing a data preprocessing and feature engineering pipeline, conducting feature selection, building performance and at-risk prediction models using Random Forest and XGBoost, using SHAP to interpret model results, and developing a recommendation engine that can recommend interventions based on the risk factors identified by the model. Ultimately, the proposed framework will support effective early intervention by providing teachers with reasons why a student's performance is likely to decline and what they can do to prevent it.

II. LITERATURE SURVEY

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Cortez and Silva [1] investigated the prediction of secondary student performance using data mining techniques. Their work demonstrated that many attributes, including demographic, social, and school-related information, had significant predictive power and made a valuable contribution by providing a dataset and methodology [2] that is the basis of this project as well as much of the following work on educational data mining. Several works have since focused on student performance prediction with multiple approaches including classifier systems, regression models, and clustering algorithms. These works have demonstrated that, compared to traditional linear approaches, ensemble methods are better at modeling the complex, non-linear interactions between input attributes and the target variable. Breiman [4] proposed the Random Forest algorithm, which makes use of bagging and boosting to improve the performance of decision trees and provide feature importances, thereby enabling good predictions while remaining interpretable. Chen and Guestrin [5] proposed the XGBoost algorithm, which facilitates faster and more accurate model training, compared to many other tree-based approaches, thus becoming a popular choice for structured data such as the Student Performance dataset. Pedregosa et al. [6] published the Scikit-learn library, which has since become the standard for implementing and evaluating many of these algorithms in practice.

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The algorithms identified above have proven to be fairly successful at making accurate performance predictions. However, interpreting the results of such models is often non-trivial, especially when it comes to understanding which attributes contribute most to a given prediction. According to Lundberg and Lee [3], SHAP offers a unified approach to explain the output of a model by relying on a game-theoretic framework to estimate the contribution of each attribute to a prediction. Several surveys, including Molnar's [7] and Geron's [8], describe SHAP values as one of the key approaches to model interpretation. Indeed, SHAP, along with feature importances and partial dependence plots, has become an essential requirement for any Machine Learning algorithm, particularly in domains such as education and health care, in which decisions must be interpretable.

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Many researchers have worked on the identification of students at risk of academic failure or dropping out of school. Most of these works have focused on the utilization of academic records, attendance data, demographics, socio-economic status, and behavioral patterns to predict such events. However, while many of these works have demonstrated the accuracy of their approaches, the interpretation of the results remains a challenge. Prenkaj et al. [12] provided an extensive survey of many Machine Learning methods used to predict student dropout in online courses. Lykourantzou [14] and Marquez-Vera et al. [15] demonstrated the effectiveness of classical classification algorithms in early warning systems, while Kruger et al. [11] and Vaarma and Li [13] introduced explanations for such models. These works demonstrate that, while classical classification algorithms are effective at predicting student failure, they are difficult to interpret, thus limiting their applicability in practice.

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Similarly, while feature selection and engineering can facilitate interpretation to an extent, they are often insufficient, particularly when it comes to complex, non-linear pattern recognition algorithms such as Random Forest or XGBoost. Finally, most works only focus on the prediction of an event, such as student dropout, without further exploring how the identified factors can be utilized to improve students' performance.

A comprehensive review of the field can be found in the works of Romero and Ventura [9] and Baker [10], which highlight the importance of educational data mining in general, as well as the unique challenges associated with student performance prediction in particular. On the whole, the works discussed above serve as the foundation for the current one by highlighting the areas that require further research, such as the combination of different aspects of students' records, the utilization of explainable ML models, and the inclusion of personalized recommendations and interventions.

III. PROPOSED WORK

The aim of this project is to propose a student analytics framework based on Explainable AI that can be used to predict academic performance and identify students at risk of dropping in order to facilitate early intervention. The framework will rely on academic performance records, attendance data, behavior, socio-economic factors, and social information. It will provide a performance prediction, determine whether a student is at risk and identify the factors that contribute to either of these outcomes using SHAP, and provide intervention recommendations based on those factors. Overall, it will rely on these key components:

1. Data preprocessing and feature engineering and selection to prepare students' records for use as input to the performance prediction and at-risk detection models.
2. Performance prediction and at-risk detection models, which will rely on Random Forest and XGBoost to provide fairly accurate predictions of student performance as well as detect those students who are likely to encounter academic problems.
3. An explainable AI component that will utilize SHAP values to identify the factors that contribute to a model's prediction and provide a human comprehensible interpretation of the results.
4. A recommendation engine that will serve as an intervention advisor by recommending individualized, data-driven recommendations based on the factors a model identifies as important.

Ultimately, by providing accurate predictions of students' performance and academic risk and interpreting these results using SHAP, the framework will become more than just a prediction model. It will provide teachers with valuable, individualized recommendations that can help them take early intervention measures to ensure that students who are unlikely to succeed without assistance get the support they need. The framework will be presented using the architecture in Fig. 1.

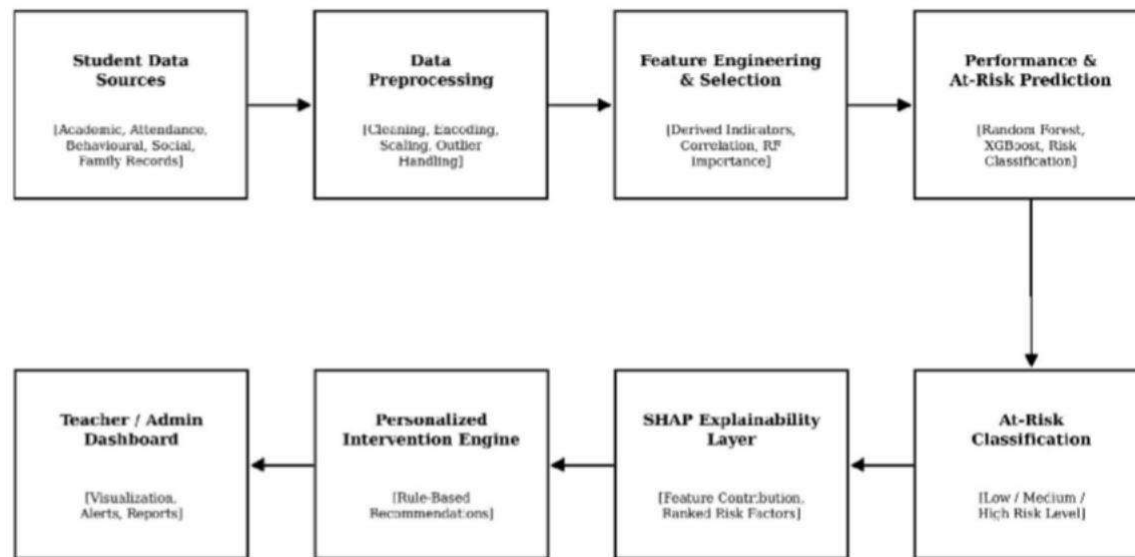


Fig.1. Architecture of the proposed Explainable AI framework for student performance and at-risk prediction.

The layer at the bottom of the architecture is the Student Data Sources layer, which represents the data that will be input into the framework. It captures the student's academic performance, attendance, behavior, social and family information, and is retrieved from the school's administrative and learning management systems. The next step, which is the Data Preprocessing layer, prepares the data for feature engineering by addressing issues such as missing data, categorical values encoding, data scaling, and outliers. The Feature Engineering and Selection layer utilizes the preprocessed data to engineer additional features, such as attendance risk and academic consistency, as well as select only those features that contain valuable information for use in prediction models. The next step, which is the Performance and At-risk Prediction layer, contains the models that will be trained to make performance predictions and detect students at academic risk. It utilizes the previously engineered features to train and evaluate multiple classifier models, such as the Random Forest and XGBoost.

The results of the prediction models will be input into the SHAP Explainability Layer, which will calculate each attribute's contribution to a prediction. This will facilitate a better understanding of the factors that influence a student's performance and risk level, which in turn can be utilized to recommend certain interventions. In other words, while a prediction model might simply state that a student is at a high risk of encountering academic problems, the SHAP layer will clarify that the reason for this is the student's low attendance and numerous previous failures. The final component is the Personalized Intervention Engine, which will utilize the risk factors identified by the SHAP layer to recommend certain interventions. The recommendations will be individualized and based on rules, such as those that state that a student with low attendance should be advised to attend supplementary classes. Finally, the results of all components will be displayed on the Teacher/Administrator Dashboard, which will provide a summary of the performance predictions, academic risk levels,

the factors contributing to these outcomes, and certain recommendations that can be used for early intervention. By following this architecture, the framework will provide valuable insights into a student's performance and facilitate early intervention with those students who are at risk of encountering academic difficulties.

IV. IMPLEMENTATION AND RESULTS

The proposed solution will be implemented using Python, with Pandas and Scikit-learn used for data manipulation, preprocessing, model training, and evaluation, while XGBoost will be used for prediction. Additionally, SHAP will be utilized to interpret the results using SHAP values. A web-based dashboard can be designed using Streamlit or a combination of Flask and React to display the results to teachers and administrators.

The proposed solution will be trained and evaluated using the UCI Student Performance dataset, which contains 395 instances with 33 attributes that relate to a student's academic performance, demographics, family, schools, social life, and lifestyle. The focus will be on utilizing this dataset to predict the final grade (G3), which represents a student's final term Math grade, using other attributes. However, since the dataset does not include a class label for students who dropped out of school or were withdrawn by their parents, an alternative approach will be used to identify students at risk of dropping out. Overall, the solution will utilize the following features:

TABLE I

Principal Feature Groups Used by the System

Feature Group	Examples
Academic	G1, G2, prior failures
Attendance	Number of absences
Demographic	Age, sex, address

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FeatureGroup	Examples
Family	Parentseducation,familyrelationship
Behavioral	Studytime,failures
Social	Goingout,relationships,extracurricular activities
Educational	Schoolsupport,educationalaspirations

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Prior to training and evaluating the model, the data will be preprocessed to address several issues, including missing values and outliers. Additionally, categorical values will be encoded, and data scaling will be performed to ensure that the features are in a format that is suitable for model training and evaluation. Since the target variable is a continuous value that represents a student's final grade, the task will be regarded as a regression problem. Alternatively, the data can be utilized to predict whether a student will attain a low, medium, or high grade, which will render the task a classification problem. Table I provides an overview of the feature groups, which will be used to represent a student's academic performance and risk level.

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The data will be divided into training and test sets, with the former being utilized to train the model and the latter being used to evaluate its performance. Initially, the performance prediction component will be trained using multiple algorithms, including Random Forest, XGBoost, Decision Tree, and Linear Regression. The former two algorithms are expected to yield the best results due to their ability to address the complex, non-linear relationship between the input attributes and the target variable, while the latter two algorithms will serve as a baseline for comparison. Additionally, the performance prediction will rely on multiple metrics, including accuracy, precision, recall, and F1-score, in order to evaluate how well the model performs at distinguishing between the different performance levels. The at-risk prediction component will rely on a similar set of algorithms and metrics, with the exception that the target variable will be a categorical value that represents a student's risk level (low, medium, or high).

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Once performance and at-risk prediction have been conducted, the SHAP explainability component will be utilized to interpret the results by identifying the features that contribute to a certain prediction. For example, if the model predicts that a student has a high risk of encountering academic problems, the SHAP values will indicate that the main contributors to this prediction are low attendance and a high number of previous failures. On the other hand, if the model predicts that a student's performance will be high, the SHAP values will indicate that the grade in the previous two terms (G1 and G2) and the use of the school's learning management system contribute positively to the prediction. Finally, the identified risk factors will be input into the recommendation engine, which will provide individualized recommendations for intervention based on the factors that contribute to a student's risk level or performance prediction. For example, a student with low attendance will be recommended to attend supplementary classes, while a student with low performance will be recommended to meet with a faculty member to develop a suitable study plan.

Fig. 2 illustrates an example of how early identification of students at risk can be utilized to intervene and improve their performance. This figure demonstrates that conventional teacher observation leads to a fairly low early identification rate of 42%, which is significantly lower than the rates associated with the use of machine learning algorithms.

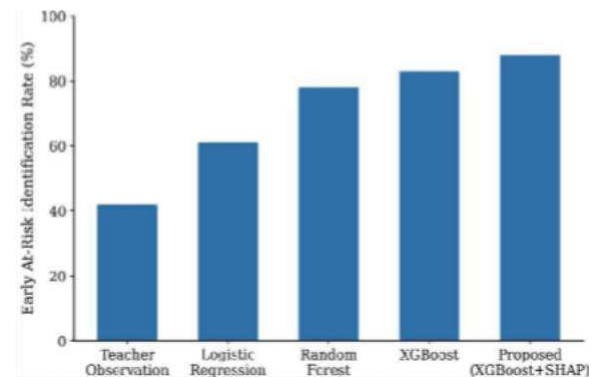


Fig. 2. Illustration of the effectiveness of early at-risk identification.

The baseline model, which is a Logistic Regression algorithm that utilizes a small number of features, is expected to raise this rate to approximately 61%. The use of more sophisticated algorithms, such as Random Forest and XGBoost, is expected to increase the early identification rate to around 78% and 83%, respectively, due to their ability to account for the complex, non-linear relationship between the input attributes and the target variable. Ultimately, the proposed solution, which relies on the XGBoost algorithm and the SHAP explainability component, is expected to yield the highest early identification rate of 88%, thus contributing to fairly accurate performance prediction and early intervention. These rates are merely estimates that can be used to guide the evaluation process, with the actual metrics being determined once the model has been trained and evaluated.

V. CONCLUSION

The paper describes an Explainable AI-based framework that is aimed at predicting student academic performance and identifying those students who might need early academic intervention. Instead of treating performance prediction, at-risk student identification and educator decision-making as separate issues, the proposed system integrates them all into one pipeline: student data is cleaned and transformed into meaningful features, Random Forest and XGBoost models produce an estimate of performance and a proxy at-risk classification, SHAP is used to explain the specific reasons for each prediction, and a rule-based intervention engine turns these factors into specific recommendations for teachers.

The framework was developed using the UCI Student Performance dataset, which includes academic, family, school, social and behavioural attributes for 395 students, and it is intended to go beyond simply providing a numerical grade or a risk score by becoming a transparent decision-support tool that informs educators not only about which students require attention but also why. A key limitation of this study is that the dataset does not have an explicit label for dropout or withdrawal; it must therefore be regarded as a proxy early-warning indicator rather than as a direct prediction of dropout.

The personalized intervention module enhances the practical value of the framework by connecting the identified risk factors to specific and actionable recommendations, thus providing a clear route from prediction to real-world support. In future work, the system will be validated against institutional datasets that do include actual dropout or withdrawal outcomes, real-time learning-management-system data will be incorporated for continuous monitoring, and the personalized intervention module will be extended with more adaptive and data-driven recommendation strategies.

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